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- 2 (a) Use standard results from the list of formulae (MF19) to find $\sum_{r=1}^n r(r+1)(r+2)$ in terms of n ,
fully factorising your answer. [3]

- (b)** Express $\frac{1}{r(r+1)(r+2)}$ in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^n \frac{1}{r(r+1)(r+2)}. \quad [5]$$

[illegible]

- (c) Deduce the value of $\sum_{r=1}^{\infty} \frac{1}{r(r+1)(r+2)}$. [1]
